

PART A — QUESTIONS

AF10-001

Question:

Solve: $12x - 9 = 63$

- A) 5
- B) 6
- C) 7
- D) 8

AF10-002

Question:

If $10n + 7 = 87$, what is n ?

- A) 6
- B) 7
- C) 8
- D) 9

AF10-003

Question:

Simplify: $4(4x - 5) + 3x$

- A) $16x - 20$
- B) $19x - 20$
- C) $19x - 5$
- D) $7x - 20$

AF10-004

Question:

A service company charges \$35 plus \$14 for each hour of work. Which expression represents the total charge for h hours?

- A) $35h + 14$
- B) $14h + 35$
- C) $49h$
- D) $35 - 14h$

AF10-005

Question:

If $f(x) = 6x - 13$, what is $f(9)$?

- A) 41
- B) 45
- C) 54
- D) 67

AF10-006

Question:

Which ordered pair satisfies the equation $y = 5x - 4$?

- A) (2, 5)
- B) (3, 11)
- C) (4, 18)
- D) (5, 24)

AF10-007

Question:

Solve: $11(x - 3) = 55$

A) 6

B) 7

C) 8

D) 9

AF10-008

Question:

A pattern begins: 7, 16, 25, 34, ... What is the next number?

A) 41

B) 42

C) 43

D) 44

AF10-009

Question:

Solve: $x/16 = 5$

A) 21

B) 64

C) 75

D) 80

AF10-010

Question:

A line has a slope of -7 and crosses the y-axis at 6. Which equation represents the line?

A) $y = 6x - 7$

B) $y = -7x + 6$

C) $y = 7x + 6$

D) $y = -6x + 7$

AF10-011

Question:

If 8 bins contain 136 parts, how many parts are in 5 bins at the same rate?

A) 75

B) 80

C) 85

D) 90

AF10-012

Question:

Solve: $60 - 9x = 24$

A) 3

B) 4

C) 5

D) 6

AF10-013

Question:

Which expression is equivalent to $32a - 14a - 9$?

- A) $18a - 9$
- B) $46a - 9$
- C) $18a + 9$
- D) $32a - 23$

AF10-014

Question:

A container starts with 210 items. Workers remove 15 items each hour. Which equation gives the number of items I left after h hours?

- A) $I = 210h - 15$
- B) $I = 15h + 210$
- C) $I = 210 - 15h$
- D) $I = 15h - 210$

AF10-015

Question:

If $y = -9x + 70$, what is y when $x = 6$?

- A) 16
- B) 24
- C) 54
- D) 124

AF10-016

Question:

Solve: $15x + 4 = 9x + 58$

- A) 7
- B) 8
- C) 9
- D) 10

AF10-017

Question:

Which value of x makes $8x - 5$ greater than 51?

- A) 6
- B) 7
- C) 8
- D) 5

AF10-018

Question:

A table shows a linear relationship.

x : 0, 1, 2, 3

y : 2, 12, 22, 32

What is the rule?

- A) $y = 2x + 10$
- B) $y = 10x + 2$
- C) $y = 12x - 10$
- D) $y = x + 2$

AF10-019

Question:

Solve: $7(x + 5) = 4x + 56$

- A) 5
- B) 6
- C) 7
- D) 8

AF10-020

Question:

If $p = -6$ and $q = 7$, what is $2p + 2q$?

- A) -26
- B) -2
- C) 2
- D) 26

AF10-021

Question:

Which point is located on the x-axis?

- A) (0, 9)
- B) (-6, 0)
- C) (-3, 5)
- D) (4, -4)

AF10-022

Question:

A worker counts 84 parts in 6 minutes. At the same rate, how many parts can the worker count in 10 minutes?

- A) 120
- B) 130
- C) 140
- D) 150

AF10-023

Question:

Solve for y : $W = 12y + 36$

- A) $y = W - 36$
- B) $y = (W - 36)/12$
- C) $y = 12(W - 36)$
- D) $y = W/12 + 36$

AF10-024

Question:

What is the slope of a line that goes through (2, 7) and (8, 31)?

- A) 3
- B) 4
- C) 5
- D) 24

AF10-025

Question:

Simplify: $22x + 6 - 13x - 18$

- A) $9x - 12$
- B) $35x - 12$
- C) $9x + 24$
- D) $35x + 24$

AF10-026

Question:

If $g(x) = x^2 + 2x$, what is $g(8)$?

- A) 64
- B) 72
- C) 80
- D) 96

AF10-027

Question:

A sequence follows the rule “multiply by 5, then subtract 4.” If the first term is 6, what is the second term?

- A) 20
- B) 24
- C) 26
- D) 30

AF10-028

Question:

Solve: $10(x - 2) + 7 = 77$

- A) 7
- B) 8
- C) 9
- D) 10

AF10-029

Question:

A line crosses the y-axis at -8 and rises 4 units for every 1 unit to the right. Which equation represents the line?

- A) $y = -8x + 4$
- B) $y = 4x - 8$
- C) $y = -4x - 8$
- D) $y = 8x + 4$

AF10-030

Question:

If $12/15 = x/45$, what is x ?

- A) 24
- B) 30
- C) 36
- D) 40

AF10-031

Question:

Solve: $-12x + 30 = -78$

- A) -9
- B) -6
- C) 6
- D) 9

AF10-032

Question:

A roll starts with 900 labels. Each box uses 45 labels. Which expression gives the number of labels left after b boxes?

- A) $900 + 45b$
- B) $900 - 45b$
- C) $45b - 900$
- D) $900 \div 45b$

AF10-033

Question:

Which equation has $x = 15$ as its solution?

- A) $2x + 8 = 36$
- B) $3x - 7 = 38$
- C) $x/5 + 4 = 8$
- D) $4x - 9 = 55$

PART B — ANSWER KEY + EXPLANATIONS

AF10-001 — Correct: B — Explanation:

Solve $12x - 9 = 63$. Add 9 to both sides: $12x = 72$. Divide by 12: $x = 6$. The most common mistake is subtracting 9 instead of adding 9.

AF10-002 — Correct: C — Explanation:

Start with $10n + 7 = 87$. Subtract 7 from both sides: $10n = 80$. Divide by 10: $n = 8$. A common mistake is stopping at 80 and forgetting to divide by 10.

AF10-003 — Correct: B — Explanation:

Distribute first: $4(4x - 5) = 16x - 20$. Then add $3x$: $16x + 3x - 20 = 19x - 20$. A common mistake is forgetting to multiply -5 by 4.

AF10-004 — Correct: B — Explanation:

The company charges \$14 for each hour, so that part is $14h$. The \$35 charge is added one time. The total charge is $14h + 35$. A common mistake is making 35 the hourly rate.

AF10-005 — Correct: A — Explanation:

Substitute 9 for x : $f(9) = 6(9) - 13 = 54 - 13 = 41$. A common mistake is adding 13 instead of subtracting it.

AF10-006 — Correct: B — Explanation:

Test the choices in $y = 5x - 4$. For $(3, 11)$, $5(3) - 4 = 15 - 4 = 11$, which matches the y -value. A common mistake is choosing a point that is close but not exact.

AF10-007 — Correct: C — Explanation:

Solve $11(x - 3) = 55$. Divide both sides by 11: $x - 3 = 5$. Add 3: $x = 8$. A common mistake is adding 3 before undoing the multiplication by 11.

AF10-008 — Correct: C — Explanation:

The pattern increases by 9 each time: 7, 16, 25, 34. Add 9 to 34: 43. A common mistake is assuming the increase is 8 because the first number is 7.

AF10-009 — Correct: D — Explanation:

$x/16 = 5$ means x divided by 16 equals 5. Multiply both sides by 16: $x = 80$. A common mistake is adding 16 and 5 instead of multiplying.

AF10-010 — Correct: B — Explanation:

Slope-intercept form is $y = mx + b$. The slope is -7 and the y-intercept is 6, so the equation is $y = -7x + 6$. A common mistake is switching the slope and y-intercept.

AF10-011 — Correct: C — Explanation:

Find the unit rate: $136 \text{ parts} \div 8 \text{ bins} = 17 \text{ parts per bin}$. For 5 bins: $17 \times 5 = 85$. A common mistake is multiplying 136 by 5 without finding the rate first.

AF10-012 — Correct: B — Explanation:

Solve $60 - 9x = 24$. Subtract 60 from both sides: $-9x = -36$. Divide by -9: $x = 4$. A common mistake is losing the negative sign after subtracting 60.

AF10-013 — Correct: A — Explanation:

Combine like terms: $32a - 14a = 18a$. The constant -9 stays the same. The expression becomes $18a - 9$. A common mistake is combining the constant with the variable terms.

AF10-014 — Correct: C — Explanation:

The container starts with 210 items. Workers remove 15 items each hour, so after h hours, $15h$ items are removed. The number left is $I = 210 - 15h$. A common mistake is adding $15h$ even though items are being removed.

AF10-015 — Correct: A — Explanation:

Substitute $x = 6$ into $y = -9x + 70$: $y = -9(6) + 70 = -54 + 70 = 16$. A common mistake is treating -9(6) as positive 54.

AF10-016 — Correct: C — Explanation:

Solve $15x + 4 = 9x + 58$. Subtract $9x$: $6x + 4 = 58$. Subtract 4: $6x = 54$. Divide by 6: $x = 9$. A common mistake is subtracting 4 from only one side.

AF10-017 — Correct: C — Explanation:

Solve $8x - 5 > 51$. Add 5: $8x > 56$. Divide by 8: $x > 7$. The only listed value greater than 7 is 8. A common mistake is choosing 7, but $8(7) - 5 = 51$, not greater than 51.

AF10-018 — Correct: B — Explanation:

The y-values increase by 10 each time x increases by 1, so the slope is 10. When $x = 0$, $y = 2$, so the y-intercept is 2. The rule is $y = 10x + 2$. A common mistake is using 2 as the slope because it is the first y-value.

AF10-019 — Correct: C — Explanation:

Distribute: $7x + 35 = 4x + 56$. Subtract $4x$: $3x + 35 = 56$. Subtract 35: $3x = 21$. Divide by 3: $x = 7$. A common mistake is forgetting to multiply 5 by 7.

AF10-020 — Correct: C — Explanation:

Substitute $p = -6$ and $q = 7$: $2p + 2q = 2(-6) + 2(7) = -12 + 14 = 2$. A common mistake is treating $2(-6)$ as positive 12.

AF10-021 — Correct: B — Explanation:

A point on the x-axis has $y = 0$. The point $(-6, 0)$ is on the x-axis. A common mistake is choosing $(0, 9)$, which is on the y-axis.

AF10-022 — Correct: C — Explanation:

84 parts in 6 minutes means $84 \div 6 = 14$ parts per minute. In 10 minutes: $14 \times 10 = 140$. A common mistake is multiplying 84 by 10 without finding the rate first.

AF10-023 — Correct: B — Explanation:

Start with $W = 12y + 36$. Subtract 36 from both sides: $W - 36 = 12y$. Divide by 12: $y = (W - 36)/12$. A common mistake is dividing W by 12 first and then subtracting 36.

AF10-024 — Correct: B — Explanation:

Slope equals change in y divided by change in x . From $(2, 7)$ to $(8, 31)$, y increases by 24 and x increases by 6. The slope is $24 \div 6 = 4$. A common mistake is using 24 as the slope without dividing by the change in x .

AF10-025 — Correct: A — Explanation:

Combine like terms: $22x - 13x = 9x$, and $6 - 18 = -12$. The simplified expression is $9x - 12$. A common mistake is writing $9x + 12$ because the negative sign before 18 is missed.

AF10-026 — Correct: C — Explanation:

Substitute 8 for x : $g(8) = 8^2 + 2(8) = 64 + 16 = 80$. A common mistake is calculating 8^2 as 16 instead of 64.

AF10-027 — Correct: C — Explanation:

Start with 6. Multiply by 5: 30. Then subtract 4: 26. A common mistake is subtracting 4 first and then multiplying.

AF10-028 — Correct: C — Explanation:

Solve $10(x - 2) + 7 = 77$. Distribute: $10x - 20 + 7 = 77$. Combine: $10x - 13 = 77$. Add 13: $10x = 90$. Divide by 10: $x = 9$. A common mistake is forgetting to combine -20 and $+7$.

AF10-029 — Correct: B — Explanation:

The line rises 4 units for every 1 unit to the right, so the slope is 4. It crosses the y-axis at -8 , so the equation is $y = 4x - 8$. A common mistake is switching the slope and intercept.

AF10-030 — Correct: C — Explanation:

Use the proportion $12/15 = x/45$. Since $15 \times 3 = 45$, multiply 12 by 3: $x = 36$. A common mistake is adding 30 to the numerator because 15 becomes 45.

AF10-031 — Correct: D — Explanation:

Solve $-12x + 30 = -78$. Subtract 30 from both sides: $-12x = -108$. Divide by -12 : $x = 9$. A common mistake is giving -9 because both sides contain negative numbers.

AF10-032 — Correct: B — Explanation:

The roll starts with 900 labels. Each box uses 45 labels, so b boxes use $45b$ labels. The number left is $900 - 45b$. A common mistake is adding $45b$ even though labels are being used.

AF10-033 — Correct: B — Explanation:

Test $x = 15$. In choice B, $3(15) - 7 = 45 - 7 = 38$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.